DS & AI LAB · GROUP 7 · MILESTONE 1 REVIEW

TerraWatch

Two satellite images of the same place. What changed, and can we say it in one honest sentence?

Priyanshu Agarwal Sarthak Saini Sujal Raj Pradhan Shoaib SSM

The problem

Today's tools stop at a **change map**: a mask of pixels that changed. An officer still has to open every flag and work out what changed.

Many flags are not real change at all: moving shadows, a different sun angle, seasons, slight misalignment.

Vision-language models can write a sentence, but they state things the image doesn't support: wrong counts, missing objects, made-up causes.

PROBLEM STATEMENT

How can two satellite images of the same place be turned into a localised, plain-language account of what changed, one that only says what the imagery supports, and holds up when the images differ for reasons that aren't real change?

Scope: urban and built-up change

In scope

- Aligned high-res RGB image pairs (~0.5 m/pixel)
- Building and road change: new, removed, groundwork
- Pixel-level change map per pair
- Short description grounded in the change map
- Robustness to lighting, colour, blur, misalignment
- Public benchmarks + human factual audit
- Local demo

Out of scope

- Real-time or nationwide monitoring
- Legality, ownership or intent
- Why a change happened
- Damage, cost or risk estimates
- Area in m² without georeference
- Validated claims on 10 m Sentinel-2, forests, disasters
- A production service

Stakeholders

Who	Role	Need	What we give them
Building and encroachment officers (e.g. BMC)	Primary	Find new structures early, verify on site	Change map + one-line description per flag
Defence Estates / cantonment boards	Primary	Catch encroachment across 62 cantonments	Same, on top of their existing image comparison
Planners and development authorities	Secondary	Track where an area is being built up	Consistent change reports
Remote-sensing researchers	Secondary	Honest evaluation of change captioning	Code, fixed splits, factual-audit protocol
Journalists, NGOs, students	Secondary	Understand imagery without GIS skills	Plain-language descriptions with stated limits

Why it matters: more imagery than people

4+ TB

of new Sentinel-2 products published every day, from one mission alone

62

cantonment boards running satellite change-detection software since 2022

2026

Mumbai's BMC approves an AI + satellite system to flag illegal construction

Both systems raise a flag, then an officer checks it. **The slow, costly part is the human check**: every false flag and every flag without context costs time.

Sources: ESA; Directorate General Defence Estates; Free Press Journal (July 2026)

How it is done today

Approach	Examples	Limitation
Classical differencing	Image difference, CVA, Otsu	Shadows and seasons count as change
Siamese CNNs	FC-EF, FC-Siam-diff, U-Net	Need pixel labels; output is only a mask
Transformers	BIT, ChangeFormer	Small gains; a plain U-Net is still near the top
Foundation / zero-shot	AnyChange (SAM), RemoteCLIP	Trail supervised models; still masks or labels
Change captioning	RSICCformer, Chg2Cap	Scored on word overlap, not on truth
Agents and VLMs	Change-Agent, ChangeChat, CCExpert	Fluent but unchecked claims; heavy to run
Operational tools	Earth Engine, Copernicus, DGDE, BMC	Flags only; a person interprets each one

Benchmarks and published baselines

Detection · LEVIR-CD

Method	F1	IoU
FC-Siam-Diff	86.31	75.92
BIT	89.31	80.68
ChangeFormer	90.40	82.48

Top methods sit within about 4 F1 points. Metrics: precision, recall, F1, IoU of the change class.

Captioning · LEVIR-MCI

Method	BLEU-4	CIDEr-D
RSICCformer	60.44	130.00
Chg2Cap	64.39	136.61
MCINet (Change-Agent)	65.95	140.29

MCINet also reaches 86.43 mIoU on the change masks. Metrics: BLEU, METEOR, ROUGE-L, CIDEr-D. None measure truth.

Sources: Bandara & Patel 2022 (ChangeFormer); Liu et al. 2024 (Change-Agent), Table II

Data: enough to train and evaluate

Dataset	What is labelled	Size	Our use
LEVIR-MCI	Masks (road, building) + 5 captions	10,077 pairs, 0.5 m	Primary: masks and captions
LEVIR-CC	Human captions	50,385 sentences	Caption baselines
LEVIR-CD	Building change	637 pairs, 1024^2	Detection benchmark
LEVIR-CD+	Building change	985 pairs	Extra detection data
S2Looking	Building change, side-looking	5,000 pairs	Robustness test
SYSU-CD	Buildings, roads, vegetation, sea	20,000 pairs	Generalisation test
SECOND	6 land-cover classes, before/after	4,662 pairs (2,968 open)	Optional: what changed
ChangeChat-105k	Instruction Q&A	~105k items	Optional: Q&A

All links checked. One private shared copy in the team Drive. Known risks: LEVIR datasets share images (leakage), labels cover only some change types, Google Earth imagery is academic-use only.

Where current solutions fall short

Detection is near-saturated

Top models are about 4 F1 apart; a plain U-Net is still a top performer.

Captioners ignore the evidence

Most never see the change map; none check the sentence against it.

Descriptions aren’t checked for truth

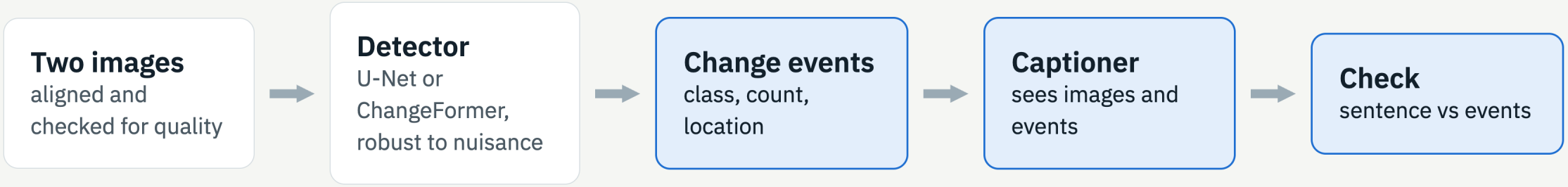
A sentence with the wrong count can still score well on BLEU or CIDEr.

Robustness is rarely reported

Scores fall sharply on side-looking S2Looking images.

Opportunity: keep a standard detector, use its map as evidence for the sentence, check the sentence, and measure how often it is right.

Our contribution: reliability, not raw accuracy



Closest precedent: Change-Agent

Trains detection and captioning together, scored with n-gram metrics. Nothing checks the sentence against the mask.

What we add

The map as explicit evidence, a claim-level check, and a human audit of factual accuracy and count error. We don’t claim the combined pipeline is new.

Measurable objectives

Objective	Metric	Success if
O1 Detect change	F1, IoU (LEVIR-CD); mIoU (LEVIR-MCI)	Within ~2 F1 of published U-Net / ChangeFormer
O2 Ignore false change	F1 drop under 5 perturbations × 3 levels	≥25% smaller drop than baseline, ≤1 F1 lost on clean data
O3 Describe truthfully	Supported-claim precision, count error, BLEU-4, CIDEr-D	+10 points supported claims vs images-only, text scores near Chg2Cap
O4 Make it usable	Latency per pair; working demo	Under 5 s per 256×256 pair on a Colab T4

Targets are hypotheses fixed in advance. If the gain is small or zero, we report that.

How we will show it works

Caption ablation

1. Images only
2. + predicted change map
3. + ground-truth map (upper bound only)
4. + consistency check

Same test pairs for every system.

Human factual audit

- ≥100 LEVIR-MCI test pairs
- 2 blind annotators per caption
- Each claim: supported, unsupported or contradicted
- Agreement: Cohen’s kappa

Honest results

- Splits and metrics fixed first
- Test set used once
- 3 seeds, bootstrap intervals
- Robustness and transfer: S2Looking, SYSU-CD

Plan and risks

M2 · Oct 29
Dataset audit, splits, leakage check

M3 · Nov 5
Architecture, end-to-end on a subset

M4 · Nov 19
Training and tuning

M5 · Nov 26
Evaluation and human audit

M6 · Dec 10
Demo and documentation

Risk	Mitigation
A dataset link stops working	Shared Drive copy; scripts and checksums in the repo
Not enough GPU time	256×256 crops, mixed precision, small backbones, Colab / Kaggle
Overlap with Change-Agent	Stated openly; our claim is the check and the audit
Labels miss real changes	Per-dataset results; unlabelled changes reported separately
Factual gain is small	Pre-registered comparison; negative result still reported

Key references

- Liu et al. 2024, Change-Agent / LEVIR-MCI, IEEE TGRS
- Liu et al. 2022, LEVIR-CC and RSICCformer, IEEE TGRS
- Chang & Ghamisi 2023, Chg2Cap
- Bandara & Patel 2022, ChangeFormer
- Chen, Qi & Shi 2021, BIT, IEEE TGRS
- Daudt et al. 2018, FC-Siam, ICIP
- Corley et al. 2024, A Change Detection Reality Check
- Ricci, Bazi & Melgani 2026, change captioning meets LLMs and VLMs, ISPRS J.
- Zheng et al. 2024, Segment Any Change, NeurIPS
- Liu et al. 2023, RemoteCLIP
- Deng, Zhou & Wu 2024, ChangeChat
- Wang et al. 2024, CCExpert
- Chen & Shi 2020, LEVIR-CD; Shen et al. 2021, S2Looking
- Shi et al. 2021, SYSU-CD; Yang et al. 2021, SECOND
- ESA, Sentinel-2; DGDE change detection; Free Press Journal, BMC (2026)

Full list with links: docs/Milestone1_Report.md, section 13